YouTube Shadowing Studio — User Manual

Interactive Web Studio for Language Shadowing, Dictation Typing & Voice Feedback

1. Overview & Core Features

YouTube Shadowing Studio allows language learners to practice listening, speaking, and typing using rich YouTube content in a streamlined split-view environment.

- **Split-View Workflow:** Watch YouTube clips on the main panel while practicing shadowing recordings, sentence typing, or taking script notes simultaneously.
- **Language Toggle (i18n):** Instantly switch the user interface between English (EN) and Korean (KO) with full state persistence.
- **Smart Timestamp Bookmarking:** Jump to specific video times (e.g., 33:00) and preserve study timestamps directly in your local browser library.
- **Voice Recorder & Visualizer:** Record your speech while shadowing clips, observe input levels on a dynamic visualizer, and review playback.
- **Real-time Typing Evaluator:** Dictate sentences with real-time character diff highlighting and accuracy score calculation.
- **Local Persistence:** All study settings and saved video cards are stored securely in your browser's `localStorage`.

2. Getting Started

1. **Opening the Application:** Open `index.html` in any modern web browser (Google Chrome, Microsoft Edge, Safari, or Firefox).
2. **Selecting Interface Language:** Click the **[EN / 한국어]** toggle button located at the top right of the header to switch to your preferred language environment.
3. **Loading a Video:**
   - Enter a YouTube video URL or 11-digit Video ID (e.g., `https://www.youtube.com/watch?v=dQw4w9WgXcQ` or `dQw4w9WgXcQ`) into the top input bar.
   - Click the **[불러오기] (Load)** button or press `Enter`.
   - Alternatively, select a video card from the saved library or click **[데모 영상 로드] (Load Demo Video)** on the player placeholder screen.

> **Verification:** The selected YouTube video is successfully loaded and ready for playback in the top player container.

3. Video Player & Navigation Controls

- **Play / Pause (`Space`):** Toggle video playback.
- **Seek -5s / +5s (`Ctrl` + ← / `Ctrl` + →):** Instantly skip backward or forward by 5 seconds.
- **Custom Time Jump & Move (`customTimeInput`):** Type a timestamp in `MM:SS` format (e.g., `33:00`) or total seconds (e.g., `1980`), then click the red **[이동] (Move)** button or press `Enter`.
- **External App / Web Launcher:** Click the external link icon button (`↗`) right next to **[이동]** to open restricted videos directly in YouTube at the specified timestamp.
- **A-B Loop Segment (`loopBtn`):** Toggle a 5-second repeating loop from your current playback position.

- **Playback Speed (`speedSelect`):** Adjust playback rate between 0.5x, 0.75x, 1.0x (default), 1.25x, and 1.5x.

4. Interactive Practice Panel

Shadowing & Voice Recording

- **Live Visualizer:** Displays real-time microphone input volume levels.
- **Start / Stop Recording (`Alt + R`):** Record your speech and listen back to generated audio preview files (.webm/.m4a).
- **Recording List:** Replay recordings or download audio clips directly to your device.

Typing Practice & Feedback

- **Target Sentence:** View target reference sentences loaded from the Script tab or custom input.
- **Real-time Diff View:** Compare input character by character: **Green** for correct text, **Red Underline** for mismatched text.
- **Accuracy Score:** Shows real-time accuracy percentage (0% to 100%).

Script & Notes Management

- **Timestamped Scripts:** Jump directly to key timestamps or select sentences for active typing practice.
- **Add Custom Lines:** Click **[+ 문장 추가]** to store new sentences matching your video position.

5. Saved Video Library & Management

The library at the bottom stores your study clips along with status indicators:

- **Status Badges:** 재생 가능 (Playable) for standard embedded clips, and 재생 불가 (Unplayable) for videos blocked by creators from third-party playback.
- **Card Actions:** Re-learn from saved timestamps, launch on YouTube, or delete cards permanently.

6. Keyboard Shortcuts

Action	Shortcut
Play / Pause Video	`Space`
Rewind 5 Seconds	`Ctrl` + ←
Forward 5 Seconds	`Ctrl` + →
Start / Stop Voice Recording	`Alt` + `R`

7. Troubleshooting & FAQs

Q1. Why does a video show "재생 불가" (Unplayable) or "This video is unavailable"?
Some YouTube creators disable embedding on external sites. Simply click the external open button (`↗`) to view the video at your saved timestamp on YouTube.

Q2. Why is my microphone not recording audio?
Ensure browser microphone permissions are allowed and that you are accessing the app over a secure `localhost` or `HTTPS` connection.